

COURSE TITLE : TEXTILE WET PROCESSING II
COURSE CODE : 4063
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Dyeing of Wool , Silk & Manmade Fibers	14
2	Dyeing of Blended material & Dyeing Machineries	14
3	Printing	14
4	Finishing	14
TOTAL		56

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Understand the dyeing of wool & silk
	2	Know the dyeing of manmade fibers
2	1	Understand the dyeing procedure of blended materials.
	2	Know the various dyeing machineries
3	1	Understand the methods & styles of printing
	2	Know the working of various printing machineries
4	1	Know the importance of finishing process and the application of different finishes to textiles

SPECIFIC OUTCOME

MODULE I DYEING OF WOOL , SILK & MAN MADE FIBERS

1.1.0 Understand the dyeing of wool & silk

1.2.0 List the various dyes used on wool and silk

- 1.2.1 Identify the chemical preparatory processes for wool and silk fibres
- 1.2.2 Explain the method of dyeing wool.
- 1.2.3 Describe the method of dyeing silk

1.3.0 Know the dyeing of manmade fibers

- 1.3.1 List the various dyes used for Man-made fibres
- 1.3.2 Explain the preparation of material, principle of dyeing and method of dyeing man-made fibres
- 1.3.3 Describe the application of disperse dyes on polyester
- 1.3.4 Explain the application of Basic dyes on Acrylic fibers

MODULE II DYEING OF BLENDED MATERIAL & DYEING MACHINERIES

2.1.0 Understand the dyeing procedure of blended materials.

- 2.1.1 Explain the Dyeing of Polyester – cotton blends with disperse and vat dyes.
- 2.1.2 Describe Dyeing of polyester – cotton blends with disperse and reactive dyes.
- 2.1.3 Explain the Dyeing of Polyester – viscose blends with disperse and vat dyes
- 2.1.4 Explain the Dyeing of Polyester – viscose blends with disperse and reactive dyes.
- 2.1.5 Explain the Dyeing of Polyester – wool blends with disperse and acid dyes.

2.2.0 Know the various dyeing machineries

- 2.2.1 Explain dyeing of loose cotton by continuous method
- 2.2.2 Illustrate package dyeing machines.
- 2.2.3 Describe the different fabric dyeing machines.
- 2.2.4 Explain Jigger, Padding mangles and Winch dyeing machine.
- 2.2.5 Explain HTHP & jet dyeing machines

MODULE III PRINTING

3.1.0 Understand the methods & styles of printing

- 3.1.1 Define printing and state the objectives of Textile printing
- 3.1.2 Differentiate Dyeing & Printing
- 3.1.3 List the ingredients used in printing & preparation of printing paste.
- 3.1.4 Describe the preparation of cloth for printing.
- 3.1.5 Differentiate between methods of printing & styles of printing.
- 3.1.6 Explain the different methods of printing.
- 3.1.7 Describe the basic concepts of different styles of printing.

3.2.0 Know the working of various printing machineries

- 3.2.1 Describe the working of flat bed screen printing machine.
- 3.2.2 Explain the features of rotary screen printing machine.
- 3.2.3 Describe the features & working of single & multicolour roller printing machines.
- 3.2.4 Illustrate the features of duplex printing machine.
- 3.2.5 Explain the working of transfer printing machine.

MODULE IV FINISHING

4.1.0 Know the importance of finishing process and the application of different finishes to textiles

- 4.1.1 State the objectives of finishing.
- 4.1.2 Classify the finishes.

- 4.1.3 Explain water mangle, Hydro extractor, Drying machine, Damping & stentering and starch finishing
- 4.1.4 Describe Calendaring & Sanforizing
- 4.1.5 State Anti-crease finishes
- 4.1.6 Illustrate water repellent and water proof finishes.
- 4.1.7 Identify mildew, moth and rot proof finishes.
- 4.1.8 Illustrate flame retardant finishes

CONTENT DETAILS

MODULE I

DYEING OF WOOL, SILK & MAN-MADE FIBRES

Various dyes used for protein fibres-Preparation of wool and silk for dyeing-Dyeing of wool with-Acid dyes-Basic dyes-Reactive Dyes-Dyeing of silk with-Acid dyes-Basic dyes-Reactive dyes-Principles and methods of dyeing of man-made fibers-Preparation of material (man-made) for dyeing-Dyeing of man-made material -Polyester with disperse dyes-Acrylic with basic dyes.

MODULE II

DYEING OF BLENDED MATERIAL AND DYEING MACHINERIES

Explain the preparation of blended material for dyeing- Principles of Dyeing & Method of Dyeing-Dyeing of polyester-cotton blends with -Disperse and vat dyes- Disperse and Reactive dyes-Dyeing of Polyester-Viscose blends with- Disperse and vat dyes-Disperse and reactive dyes-Dyeing of Polyester - wool blends with disperse and acid dyes-Classification of dyeing machinery -Loose cotton dyeing machines-Yarn package dyeing machines-Cop dyeing machine-Cheese dyeing machine-Beam warp dyeing machine-Fabric Dyeing machines-Jigger dyeing machine-Winch dyeing machine-Padding mangles-HTHP jet dyeing machines-HTHP Jigger-HTHP Beam Dyeing-Thermosol process.

MODULE III

TEXTILE PRINTING

Definition and objectives of printing-Differentiate between dyeing & printing-Ingredients used in printing & preparation of print paste-Preparation of cloth for printing-After treatments given to printed fabrics-Differentiate methods of printing & styles of printing- Study of different methods of printing - Block printing- Stencil Printing-Screen Printing-Automatic flat bed screen printing-Rotary screen printing-Roller printing-Transfer printing-Study of basic concepts of following styles of printing-Direct, Discharge, Resist and Dyed styles-Batik style and dyes used for batik style- Tie and dyeing- Foam printing- Flock printing- Damask printing-Kalamkari printing.

MODULE IV

FINISHING

Objectives of Finishing- Classification of finishing- Describe Water Mangles- Three and Six bowl mangles- Hydro extraction- Roller and Centrifuge type- Drying machines – Blanket drying & Buti hot air drying machines-Damping machines – Brush and spray-Objectives of stentering -Description of pin & clip stenter-Hot air stenter-Starch finishing-Objectives-Various ingredients-Method of application-Calendarizing-Objectives-Different types of calenders-Beetling-Sanforizing-Objectives-Principle and mechanism of compressive shrinkage-Pre-shrinkage process-Description of sanforizing machines- Anti-crease finish- Objectives-Types of anti crease finishes-Different method of anti crease finishing-Study of Water repellent & water proof finishes-Study of mildew, moth and rot proof finishes-Study of flame retardant finishes-Defects in finishing & remedies

REFERENCE BOOKS

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|----|------------------|---|
| 1. | V.A.Shenai | Technology of dyeing |
| 2. | R.S.Prayag | Dyeing of polyester and polyester blends |
| 3. | V.A.Shenai | Technology of Printing |
| 4. | J.T.Marsh | Textile Finishing |
| 5. | E.R Trotman | Dyeing & Chemical Technology of textiles |
| 6. | D.G.Kale | Principles of Textile Printing |
| 7. | Srivastave | Recent processes in Textile Bleaching, Dyeing & Finishing |
| 8. | R.R.Chakravarthy | A glimpses on the Chemical Technology of Textile Fibres |
| 9. | R.S.Prayag | Textile Printing |